

Writing Logbook — Assignment 01

CSP529 Technical Writing & Publishing

Can AI Write Your Technical Report?

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1. Assignment Overview

Item	Detail
Practical Title	CSP529 Technical Writing & Publishing: Can AI Write Your Technical Report?
Problem Statement	Develop a Drone-based Crop Disease Detection System
Group Members	Gaurav Acharya (MT26MCS024), Prasanna Thulkar (MT26MCS028), Sandeep (MT26MCS032)
Guide	Dr. A. S. Mokhade
Date of Completion	September 2026

2. Objectives & Learning Outcomes

The primary objective of this assignment was to critically evaluate whether AI tools can adequately write technical reports by comparing human-written and AI-generated problem definitions for the same engineering system. The expected learning outcomes included:

- Formulating a precise engineering problem definition with stakeholder analysis.
- Generating an AI version of the same problem definition using a language model.
- Systematically comparing human vs. AI outputs across multiple evaluation metrics.
- Identifying weaknesses in AI-generated text that appear superficially convincing.
- Developing critical thinking about AI limitations in technical writing.

3. Work Performed — Stage-by-Stage

3.1. Stages 1 & 2: Problem Definition — Student Version

We independently formulated the problem definition for a Drone-based Crop Disease Detection (CDD) System, addressing:

- **Problem:** Traditional manual crop inspection is time-consuming, labour-intensive, and prone to missing infected patches due to limited human field-of-view.
- **Root Cause:** Human physical limitations — small observation area ($< 2\%$ ground cover per sweep) and cognitive fatigue over extended acreages.
- **Stakeholders:** Local farmers, agriculture-based industries (grain processors), animal husbandry (husk/fodder suppliers).
- **Assumptions:** System would be cheaper, more efficient, and achieve high-precision detection.
- **Missing Information:** Environmental conditions (temperature, wind speed), power supply infrastructure, surrounding flora/fauna baseline data.

3.2. Stage 3: AI-Generated Version

We prompted an AI model to produce the same problem definition. Key observations:

- AI used authoritative buzzwords (“autonomous,” “machine learning,” “optimize”) but provided no quantifiable metrics.
- AI assumed continuous battery power, clear weather, and universal high-speed internet — all unrealistic in rural agricultural settings.
- AI’s missing information focused on hardware specs (battery mAh, payload weight) but ignored environmental and human factors.

3.3. Stage 4: Comparison Table

We built a structured comparison across five evaluation metrics (Problem Definition, Reason for Existence, Stakeholders, Assumptions, Missing Information), revealing that:

- The student version was grounded in practical agricultural realities.
- The AI version was technically polished but disconnected from field constraints.

3.4. Stage 5: Cross-Group Insights

After exchanging work with peer groups, we identified additional practical challenges:

- Legacy infrastructure integration difficulties.
- Unavailable data: historical leakage records, real-time pipeline conditions.
- Ethical issues: algorithmic bias, surveillance concerns, accountability for failures.
- Unrealistic assumptions: perfect system accuracy, purely trend-based demand prediction.
- Failure scenarios: sensor failures requiring fail-safe protocols; flood risks causing hardware damage; internet outages requiring offline fallback modes.

3.5. Stage 6: Revised Technical Document

We rewrote the problem definition with engineering precision:

- Quantified the problem: 40 man-hours per 100-hectare inspection cycle, 7% miss rate.
- Specified concrete assumptions: capital expenditure recovery within 2 harvest cycles, spectral classification accuracy > 92%.
- Identified precise missing metrics: wind shear velocity thresholds, localized baseline reflectance spectra for native weed flora.

3.6. Stage 7: AI Weakness Analysis

We identified three critical engineering flaws in the AI output:

1. **Ignored Physical Constraints:** “Continuous battery power” dismisses payload-weight vs. discharge-rate trade-offs.
2. **Disconnected from Field Reality:** “Clear weather” and “universal internet” ignore rural microclimates and telemetry dropouts.
3. **Zero Measurability:** “Optimize farm management” provides no metrics, parameters, or thresholds for engineering evaluation.

4. Individual Reflections

4.1. Gaurav Acharya (MT26MCS024)

- **AI cannot:** Conduct physical reality checks (test camera lens behaviour in morning dew), calculate real-world degradation (battery drain from wind shear), or gather human intelligence (farmer interviews about interface literacy).

- **AI excels at:** Formatting raw telemetry data into tables, summarizing lengthy regulatory manuals, and correcting passive voice in hardware descriptions.
- **Skill to develop:** Field-testing validation and observational rigor — bridging theoretical AI text with physical constraints.

4.2. Sandeep (MT26MCS032)

- **AI cannot:** Evaluate economic viability for local farmers, make strategic accuracy-vs-cost trade-offs, or navigate privacy/liability concerns for aerial surveillance.
- **AI excels at:** Tailoring communication for different audiences, drafting executive summaries, and brainstorming risk matrices.
- **Skill to develop:** Strategic cross-functional analysis — connecting engineering constraints with business viability and stakeholder needs.

5. Challenges Encountered

1. **Quantifying vague problems:** Translating “crops get diseased” into measurable parameters (7% miss rate, 40 man-hours) required significant domain research.
2. **Resisting AI polish:** The AI’s grammatically perfect output was initially convincing; recognizing its superficiality required deliberate critical reading.
3. **Cross-group coordination:** Aligning vocabulary and evaluation criteria across different problem domains during Stage 5 was challenging.

6. Key Takeaways

1. AI-generated text can appear authoritative while lacking engineering substance — “sounds right” $\neq$ “is right.”
2. Every technical claim must be grounded in measurable, verifiable parameters.
3. Human domain expertise, stakeholder empathy, and physical-world awareness remain irreplaceable in technical writing.
4. The iterative process of writing $\rightarrow$ comparing $\rightarrow$ revising significantly strengthened the quality of our problem definition.